Review changelog: batch p11 (Phase 11, lessons 01-24)

Reviewer date: 2026-10-22. Sandbox: react/react-dom 19.3.1, TS 5.9, Vitest 3.2 (Vitest 5.0.5 used in one scratch project), jsdom, RTL 16, TanStack Query 5.104, zod 4.6. Scratch files: `/Users/todorbra/Projects/ReactLearning/tools/sandbox/scratch/p11/` (and `.../scratch/p11/{a11y,inert,uses,r,fc}`).

Legend: confirmed / corrected / hedged / unresolved. Roughly 119 soft hedge markers (warning sign) remain in the phase, mostly generic "evaluate/verify for your org/jurisdiction/vendor" caveats on non-verifiable judgement claims (legal, vendor maturity, browser-support rules of thumb, interview content in 23). I resolved every marker attached to a tool/config/API/date/version claim that could be checked, and left the rest as intentionally hedged.

Corrections of substance (things that were wrong)

Lesson	Wrong claim	Evidence	Fix
13 (5.3)	dependency-cruiser rule `to.path: '^apps/(?!\\1/)'` "backreference trick" and `only-public-entrypoints` with `pathNot: '^packages/([^/]+)/'`	Ran depcruise 18.7: rule 1 flagged legitimate same-app imports (regex backreference does not refer to the `from` group); rule 3 never fired (pathNot excluded every package path)	Rewritten with native `$1` / `$2` capture-group syntax; both rules now behave correctly (tested)
03 (5.4)	Codemod dropped a `// keep me` comment above the replaced import, contradicting its own fixture list	Ran jscodeshift 17.4 on fixtures	Added `next.comments = p.node.comments`; added `export const parser = "tsx"`
03 (5.4)	`defineTest(__dirname, ...)` test layout	testUtils resolves transform at `<dirName>/../name` and fixtures at `../__testfixtures__`; requires global describe/it, `require`s transform (TS fails under Vitest); unchanged-file fixture must be EMPTY	Layout moved to `__tests__/`, caveats documented, `runInlineTest` alternative (passes under Vitest)
07, 23	DTCG tokens as `"$value": "#1d4ed8"` / `"8px"` with no `$type` and "draft standard"	designtokens.org: first stable version **2025.10** (2025-10-28); color value is an object (`colorSpace`, `components`, `hex`), dimension `{value, unit}` (px/rem)	Examples rewritten to 2025.10 form; status corrected; note that older strings/tools exist; Style Dictionary docs say 2025.10 not fully supported yet
07 (C2)	token checker script would iterate the characters of a group-level `$type` string and mis-handle object values	Ran the script against the new tokens (found "missing token" false errors from stringified object values matching the alias regex)	Script skips `$`-keys and treats non-string values as literals; re-run detects real missing/parity errors

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09	`check-budgets.ts` app-entry heuristic `k.includes('main')` never matched	Real Vite 8.3 build: HTML entry key is `index.html`	Heuristic fixed; manifest shape verified (`.vite/manifest.json`, `imports`, `dynamicImports`, `isEntry`)
09	size-limit snippet silently needs a plugin; default compression is Brotli	size-limit README v14	Added `@size-limit/file` requirement and Brotli/gzip note
09	LHCI "median of runs" implied	LHCI docs: `assert` default `aggregationMethod: "optimistic"`	Documented; default 3 runs, `numberOfRuns`
10	Fake-timer debounce test (user-event + `vi.advanceTimersByTime`)	**Hangs under Vitest** unless `globalThis.jest = { advanceTimersByTime: vi.advanceTimersByTime }` (RTL async wrapper) - reproduced and fixed with shim	Added caveat + shim
10	RTL auto-cleanup	With Vitest `globals:false`, DOM leaks between tests ("multiple elements")	Documented `afterEach(cleanup)`; jest-dom import `@testing-library/jest-dom/vitest`
10	`toHaveBeenCalledWith({cart}, {})` on a function component	React 19 passes `undefined` as 2nd arg (React 18: `{}`)	Corrected
11	`showModal()` "throws on an already-open dialog"	HTML spec: no-op if already open as modal; throws only if open non-modally / not connected	Corrected
11	WCAG 2.2 listing incomplete/uncertain	W3C TR: 9 new SC, 4.1.1 removed; Rec 2023-10 re-published 2024-12-12	Added SC numbers and levels
17	ESLint flat config would crash: `recommendedTypeChecked` without `parserOptions.projectService`; `tseslint.config` deprecated	Ran ESLint 9.39 + typescript-eslint 8.72: "requires type information" error	Config fixed with `defineConfig` and `projectService` (tested)
22	CI canary uses `react@rc`	`npm view react dist-tags`: `rc` = **19.0.0-rc.1** (older than latest 19.3.1)	Switched to `canary` and documented the pitfall; `eslint-plugin-deprecation` -> `@typescript-eslint/no-deprecated`
02 (5.5)	fast-check round-trip property fails once the implementation canonicalises `status` order	Ran fast-check 4 + Vitest: counterexample `["open","cancelled"]`	Generator now emits canonical order; caveat rewritten

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16, 17	DORA described as four metrics with vague "verify"	dora.dev: **five** metrics (change lead time, deployment frequency, failed deployment recovery time, change fail rate, deployment rework rate)	Updated
20	`<ViewTransition>` "experimental"; Compiler/useEffectEvent/Activity statuses hedged	react.dev blog: View Transitions + Fragment Refs **stable in React 19.3 (2026-09-09)**; Compiler 1.0 2025-10-07; useEffectEvent/Activity stable in 19.2	Table, example comments, answer key (C2) updated
05, 06, 19	"React 19 added `onRecoverableError`"	`onRecoverableError` is from React 18; 19 added `onCaughtError` / `onUncaughtError`	Corrected; payload shapes run-tested
15	Framework list/status hedged	react.dev: Next.js App Router, React Router v7, Expo; TanStack Start labelled Beta on react.dev but `@tanstack/react-start` is 1.x on npm; Next 16 caching via `"use cache"` / `cacheComponents`	Updated with dated statements
21	Many "search seed" names from memory	Checked each against facebook/react `main` and installed react-dom 19.3.1 dev bundle; file locations added. `tryToClaimNextHydratableInstance` exists on main but not in the 19.3.1 bundle; scheduler yields via `setImmediate` first, then `MessageChannel`; compiler lives in top-level `compiler/`	Table rewritten with verified names/files

Per lesson

- **01**: ESLint/ADR script statements not tool-specific; `<dialog>` support remark hedged (browser matrix). Confirmed: nothing wrong found; remaining hedges are judgement (unresolved by design).
- **02**: 5.5 corrected (above). Confirmed fast-check `requiredKeys`, `subarray`, types compile (`tsc --strict`).
- **03**: codemod/testing corrected; `.git-blame-ignore-revs` claim confirmed for GitHub (hedged for other hosts); jscodeshift 17.4.0 and ts-morph 28 current. Ratchet script read, not executed (**unresolved**: first run crashes if baseline file missing; not changed).
- **04**: ref-as-prop, forwardRef "legacy, will be deprecated" confirmed on react.dev; compiler 1.0 status; `Math.round(-2.5) === -2` confirmed by node.
- **05**: ESLint flat `no-restricted-imports` run (matches critique); depcruise config run (`$1` works); `onCaughtError` etc. corrected; MF `singleton` confirmed; TS `exports` resolution wording corrected (node16/nodenext/bundler honour it). Remaining hedges: CSP/sandbox/legal vendor remarks (sandbox `allow-same-origin` + `allow-scripts` warning is correct).
- **06**: root error options run-tested; `<Ctx value>` / `.Provider` both verified. Other hedges (legal/CDN) unresolved by design.
- **07**: tokens/DTCG corrected; contrast function run (`#000/#fff` 21, `#767676/#fff` 4.54, lesson pair 16.98); `forwardRef` wording; Changesets cross-link. Remaining hedges are bundler/RSC behaviour that vary (unresolved).

- **08**: TanStack Query v5 defaults confirmed in `@tanstack/query-core` source (`staleTime` 0, `gcTime` 3e5 ms browser / Infinity server, `cacheTime` renamed); the full optimistic-update snippet run with RTL (optimistic then rollback) and typechecked. `<Ctx value>` verified. useSyncExternalStore per-symbol predict consistent with the object-identity behaviour verified in lesson 21.
- **09**: size-limit, LHCI, web-vitals 6.2.3 exports, thresholds (`LCPThresholds` [2500,4000], `INPThresholds` [200,500], `CLSThresholds` [0.1,0.25]) confirmed; manifest shape verified by real Vite build; budget script run (passes/fails as designed). FID->INP March 2024 confirmed from memory + lesson already matches web.dev (not re-fetched: **unresolved but low risk**).
- **10**: act import (react-dom/test-utils still exports a deprecated forwarding `act`), Vitest/RTL fake-timer hang, cleanup, 2nd arg, `toHaveBeenCalledExactlyOnceWith` (Vitest 3) all run. Race-condition test passes. react-router `MemoryRouter` kept (hedged).
- **11**: vitest-axe 0.1.0 run on Vitest 3 + jsdom (flags missing alt, passes good markup; not happy-dom per its README); WCAG 2.2 SC/levels checked against W3C TR; EAA date 28 June 2025 confirmed (qualified); `inert` boolean verified by SSR on react-dom 19.3.1; `showModal` spec fix.
- **12**: `@module-federation/enhanced@2.9.2` exports `ModuleFederationPlugin` from `/webpack` (required in node); shared options `singleton` / `requiredVersion` / `eager` confirmed in module-federation.io; type sharing (`@mf-types.zip`, `dts` option) confirmed; `single-spa-react@6.0.2` `ReactDOMClient` option and auto `createRoot` confirmed from its published source (`update` lifecycle also exists). Remaining hedges: SSR story per plugin, `strictVersion` (not re-verified; **unresolved**).
- **13**: Turborepo 2.11.10: `tasks` accepted, `pipeline` rejected with the exact message, `--filter="...[origin/main]"` valid, `$schema` URL moved to turborepo.com; dependency-cruiser rules fixed; Changesets config keys/ `status --since` verified from docs (CLI run not completed: interactive prompt hung, **partly unresolved**; docs recommend the changeset bot instead of the CI check); affected-packages script executed against a 3-package toy repo (direct, transitive, global-input cases correct). Nx: only named in prose (Nx 24 on npm), no claims to verify.
- **14**: flag-port snippet reviewed; `use(Context)` / `<Ctx value>` OK; OpenFeature maturity hedge kept (**unresolved**, judgement).
- **15**: see table. **16/17/18**: DORA fixed; SPACE paper citation; 17 ESLint fixed; `react/no-array-index-key` exists.
- **19**: createLatest executed (stale ignored, abort handled); zod 4.6 snippet run; root options corrected; Activity stable.
- **20**: see table; `react-compiler-healthcheck` exists on npm 1.0.0 but is not prominent in react.dev docs (stated); `@vitejs/plugin-rsc` 0.5.x (early).
- **21**: all function names checked; set-state/object-selector predictions run: primitive selector 1 render before and after unrelated update; object selector logs "The result of getSnapshot should be cached to avoid an infinite loop" then throws "Maximum update depth exceeded"; StrictMode ref counter = 2 after mount. Repo moved: `github.com/facebook/react` now 301-redirects to `github.com/react/react` (react.dev links point there); URLs left, redirect works.
- **22**: see table; React history rows consistent with react.dev (18.3 warnings, 19 removals); codemod ran + typechecked; Renovate keys not run through the validator (**unresolved**, hedged with the validator command).
- **23**: DTCG sketch fixed; thresholds consistent with 09; other content is interview-answer material with hedges kept (browser-support remarks such as `getCoalescedEvents`, `content-visibility`: **unresolved**, not verified).
- **24 (assessment)**: executed R1-R6 in jsdom with `createRoot` and real clicks on react-dom 19.3.1: R1 `["render 0","handler 0","render 2"]` /shows 2; (c) `handler 0, render 1`; R2 A keeps `x` and same DOM nodes, B and C reset and recreate; R3 mount and update orders exactly as keyed (`child layout cleanup 1` before `child layout 2` before `child cleanup 1` before `child effect 2`); R4 stays 1; R5

`child,child` / `child` / `child,child`; R6 `render #1, render #2 a=1 b=1, after await, render #3`. React 17 claim for R6 run under react@17 (render sync before `after await`). All answer-key statements agree with these and with Phase 2/3/9 behaviour (batching, effect order, Strict Mode double render, bailout). Corrected: `useEffectEvent` (stable since 19.2, restrictions, unstable identity) and `<Activity>` marker text; removed stale version flags.

Unresolved / not verified

- Changesets CLI end-to-end run (hung on interactive init); config keys verified from docs only.
- Module Federation `strictVersion` semantics, per-plugin SSR support.
- Renovate config validity; GitLab/Bitbucket blame-ignore behaviour; Vite RSC plugin maturity beyond version number.
- Browser-support claims in 23 and 08 (BroadcastChannel/Web Locks, `getCoalescedEvents`, `content-visibility`, OffscreenCanvas).
- European Accessibility Act scope details beyond the 28 June 2025 application date (legal; left hedged).
- Nx claims: none made beyond naming.
- React 19.3 "Fragment Refs", `browser()` and Trusted Types additions are new and not referenced by lessons; no change made.
- Newer majors noted by the brief (TS 7, Vitest 5, MSW 3): lessons in this phase do not pin these tools' versions except fast-check/Vitest 3 run notes, so no rewrite was needed. react-router is at 8.x on npm (lesson 15 notes this); Next.js 16.